Test Series No.

1.8

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 03/03/2026

SYLLABUS

Physics:	12th Full Syllabus
Chemistry:	12th Full Syllabus
Botany:	12th Full Syllabus
Zoology:	12th Full Syllabus

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

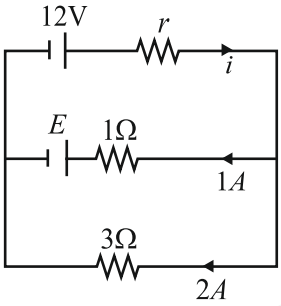
Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. In the circuit shown in figure, a 12 V battery with unknown internal resistance r is connected to another battery with unknown emf E and internal resistance 1 Ω and to a resistance of 3 Ω carrying a current of 2 A. The current through the rechargeable battery is 1 A in the direction shown. The values of internal resistance r and the emf E respectively are;

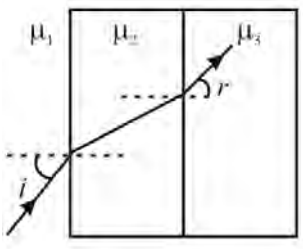


(1) 2 Ω, 5 V (2) 2 Ω, 4 V
(3) 3 Ω, 6 V (4) 3 Ω, 5 V

2. A frog is levitated in a magnetic field produced by current carrying in a vertical solenoid placed below it. Which of the following behaviour is shown by the frog?

(1) Diamagnetic (2) Paramagnetic
(3) Ferromagnetic (4) Ferrimagnetic

3. In the figure shown, value of $\frac{\sin i}{\sin r}$ is equal to: (symbols have usual meanings)



(1) $\frac{\mu_2^2}{\mu_3\mu_1}$ (2) $\frac{\mu_3}{\mu_1}$
(3) $\frac{\mu_3\mu_1}{\mu_2^2}$ (4) $\frac{\mu_1}{\mu_3}$

4. The current in a wire increases linearly from 0 to I_0 over a time interval t. What is the mean value of the current during this interval?

(1) I_0 (2) $\frac{I_0}{\sqrt{2}}$
(3) $\frac{I_0}{2}$ (4) $2I_0$

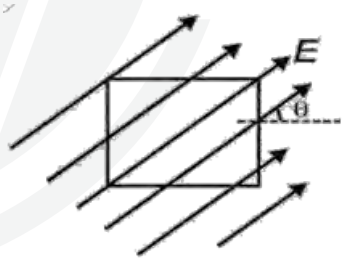
5. A long magnetic needle of length $2L$, magnetic moment M and pole strength m is broken into two at the mid point transversely (perpendicular to length). The magnetic moment and pole strength of each piece will be:

(1) $\frac{M}{2}, \frac{m}{2}$
(2) $M, \frac{m}{2}$
(3) $\frac{M}{2}, m$
(4) M, m

6. If the highest frequency of the Lyman series is ν_L, then the highest frequency of the Pfund series is:

(1) $\frac{\nu_L}{16}$ (2) $\frac{\nu_L}{25}$
(3) $25\nu_L$ (4) $16\nu_L$

7. A square surface of side L in the plane of the paper is placed in a uniform electric field E acting along the same plane at an angle θ with the horizontal side of the square as shown in figure. The electric flux linked to the surface, is:



(1) Zero (2) EL^2
(3) $EL^2\cos\theta$ (4) $EL^2\sin\theta$

8. The electric field of an electromagnetic wave is given by $\vec{E} = 60\sin\left(1.5\times10^{11}t - 500x\right)\hat{k}$ V/m. (where x is in metre and t is in seconds). Which of the following statements is **correct** regarding this wave?

(1) The wavelength is 1.5 cm.
(2) The wave propagates in the negative x - direction.
(3) The amplitude of the magnetic field is 2×10^{-7} T.
(4) The frequency of the wave is 500 Hz

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9. A parallel plate capacitor of capacitance 2μF is being charged by a voltage source whose potential is changing at the rate of 3 V/s. Given below are two statements:

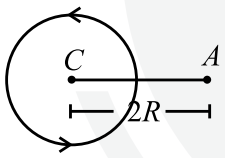
Statement I: The conduction current through the connecting wire is 6μA.

Statement II: The displacement current through the plates of the capacitor during charging is 2μA.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

10. An α particle is moving along a circle of radius R with a constant angular velocity ω. Point A lies in the same plane at a distance $2R$ from the center. Point A records magnetic field produced by α particle. If the minimum time interval between two successive times at which A records zero magnetic field is 't', the angular speed ω, in terms of t is:



(1) $\frac{2\pi}{t}$ (2) $\frac{2\pi}{3t}$
(3) $\frac{\pi}{3t}$ (4) $\frac{\pi}{t}$

11. The electric field in a certain region is acting radially outwards, and its magnitude is given by $E = Ar$ where A is constant. Total charge contained in a sphere of radius 'a' centred at the origin of the field, will be given by:

(1) $\epsilon_0 Aa^3$ (2) $4\pi\epsilon_0 Aa^2$
(3) $A\epsilon_0 a^2$ (4) $4\pi\epsilon_0 Aa^3$

12. A boy is standing midway between a vertical wall of height $H = 4.5\text{m}$ and a parallel plane mirror. The minimum height of the mirror required for the boy to see the full height of the wall behind him is:

(1) 2.25m (2) 1.5m
(3) 0.75m (4) 3.0m

13. R.M.S. value of current $i = 3 + 4\sin\left(\omega t - \frac{\pi}{3}\right)$ A is:

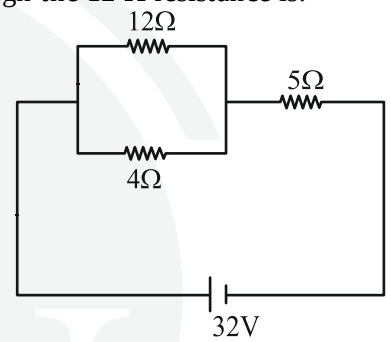
(1) 5 A (2) $\sqrt{17}$ A
(3) $\frac{5}{\sqrt{2}}$ A (4) $\frac{7}{\sqrt{2}}$ A

14. A positive point charge Q is placed at the origin $(0,0)$. Consider two points $A(3\text{m}, 4\text{m})$ and $B(5\text{m}, 0)$.

The potential difference $V_A - V_B$ is:

(1) Zero (2) $\frac{kQ}{5}$
(3) $\frac{kQ}{25}$ (4) $-\frac{kQ}{5}$

15. For the circuit shown in the figure, the current through the 12 Ω resistance is:



(1) 4A (2) 3A
(3) 6A (4) 1A

16. A galvanometer with a full-scale deflection current of 2 mA is converted into a voltmeter of range 0 to 10 V by connecting a resistance of 4950Ω in series. The resistance of the galvanometer is:

(1) 40Ω (2) 100Ω
(3) 25Ω (4) 50Ω

17. The wavelength of the third line of the Balmer series of a hydrogen like atom is equal to 1085 Å. The binding energy of an electron in the ground state of this atom is: (Take $hc = 12375$ eVÅ)

(1) 45.8 eV (2) 54.3 eV
(3) 13.6 eV (4) 23.8 eV

18. Which of the following statements regarding magnetic field lines is physically **correct**?

(1) Magnetic field lines can intersect.
(2) The net magnetic flux through any closed gaussian surface is always zero.
(3) Magnetic field lines always start from the South pole and end at the North pole outside the bar magnet.
(4) Magnetic field lines are conservative in nature.

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19. A magnet is taken towards a conducting ring in such a way that a constant current of 10 mA is induced in it. The total resistance of the ring is 0.5Ω. In 5 s, the magnetic flux through the ring changes by:

(1) 0.25 mWb (2) 25 mWb

(3) 50 mWb (4) 15 mWb

20. An intrinsic semiconductor has an intrinsic carrier concentration of 1.5×10^{16} m^{-3}. It is doped with a trivalent impurity such that the hole concentration becomes 4.5×10^{22} m^{-3}. The concentration of conduction electrons in the doped semiconductor is:

(1) 5×10^{9} m^{-3} (2) 3×10^{10} m^{-3}

(3) 2.25×10^{16} m^{-3} (4) 0.5×10^{8} m^{-3}

21. The distance between an object and its upright image is 20 cm. If the magnification is 0.5, then the focal length of the lens that is being used to form the image is:

(1) 20 cm (2) –20 cm

(3) 40 cm (4) –40 cm

22. An air capacitor has a capacitance of 4 pF. If the separation between the plates is halved and the space is completely filled with a dielectric medium of dielectric constant K, the new capacitance becomes 40 pF. The value of K is:

(1) 2 (2) 10

(3) 5 (4) 8

23. A beam of light consisting of two wavelengths 6500 A and 5200 A is used to obtain interference fringes in YDSE. The distance between slits in 2 mm and the distance of the screen from slits is 120 cm. What is the least distance from central maximum where the bright fringe due to both wavelengths coincides?

(1) 0.156 cm (2) 0.312 cm

(3) 0.078 cm (4) 0.468 cm

24. Two point charges $+9q$ and $-q$ are separated by a distance of 16 cm. The electric field intensity is zero at a point on the line joining the two charges, located:

(1) 4 cm from $+9q$ between the charges

(2) 8 cm from $-q$, away from the charge $+9q$

(3) 12 cm from $+9q$, away from the charge $-q$

(4) 4 cm from $-q$ between the charges

25. A photosensitive metallic surface is illuminated alternately with light of wavelength 3100A and 6200A. It is observed that maximum speeds of the photoelectrons in two cases are in ratio $\frac{v_{3100\text{Å}}}{v_{6200\text{Å}}}=\frac{2}{1}$.

The work function of the metal is: ($hc=12400\text{eV Å}$)

(1) 1 eV (2) 2 eV

(3) $\frac{4}{3}$eV (4) $\frac{2}{3}$eV

26. Nuclear energy is released in a fusion reaction, since binding energy per nucleon is:

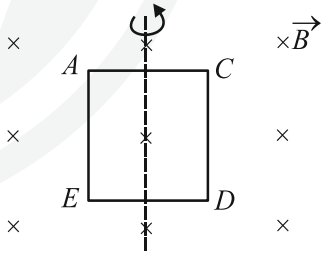
(1) Smaller for product nuclei than for fusing lighter nuclei

(2) Same for product nuclei as for fusing lighter nuclei

(3) Larger for product nuclei than for fusing lighter nuclei

(4) Constant for all nuclei regardless of mass number

27. A square loop $ACDE$ of area 10cm^2 and resistance 10 Ω is initially placed with its plane perpendicular to external magnetic field as shown. It is then rotated in a magnetic field $B=2\text{T}$ through 180° in 0.01 s. The magnitude of average value of induced emf in this interval is:



(1) 0.2 V (2) 0.4 V

(3) 0.8 V (4) 1.0 V

28. If a variable frequency ac source connected to a capacitor then with decrease in frequency, the displacement current will

(1) increase

(2) decrease

(3) remain constant

(4) first decrease then increase

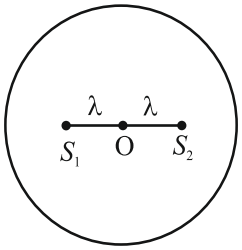
29. Which of the following is not equal to 1 in Boolean algebra?

(1) $A+1$ (2) $A+\bar{A}$

(3) $\overline{A\cdot\bar{A}}$ (4) $A\cdot\bar{A}$

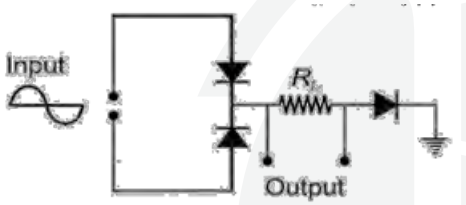
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30. Two coherent light sources S_1 and S_2 are kept at equal distances λ from the center of circular screen in the same plane of the circle as shown in figure. In one complete revolution on the circular path, number of maxima observed in circular screen are: (λ = wavelength of light used)



(1) 4 (2) 6
(3) 8 (4) 12

31. In the circuit shown, the input waveform is given. Which of the following correctly gives the output waveform across R_L ? (Assume ideal diode)



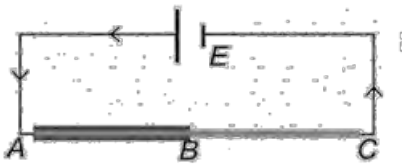
(1)  (2) 
(3)  (4) 

32. Two polaroids P_1 and P_2 are placed in a crossed position (axes perpendicular). Unpolarized light of intensity I_0 is incident on P_1. A third polaroid P_3 is inserted between them. Match the angle (θ) of P_3's axis with respect to P_1's axis (**List-I**) with the final intensity transmitted through P_2 (**List-II**).

	List-I (Angle θ)		**List-II (Final Intensity)**
(A)	45°	(I)	$\frac{I_0}{16}$
(B)	22.5°	(II)	$\frac{3I_0}{32}$
(C)	30°	(III)	$\frac{I_0}{8}$
(D)	0°	(IV)	0

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-II, D-IV
(2) A-II, B-IV, C-I, D-III
(3) A-I, B-II, C-III, D-IV
(4) A-III, B-II, C-I, D-IV

33. In the following diagram, the lengths of wires *AB* and *BC* are equal, but the radius of wire *AB* is double that of *BC*. The ratio of potential gradient along wires *AB* and *BC* will be (wires are made of same material)



(1) 4:1 (2) 1:4
(3) 2:1 (4) 1:1

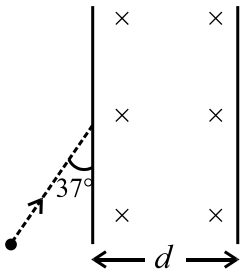
34. A ring of radius R has charges q and $3q$ distributed uniformly on its two half parts. The electric potential at a point on its axis at a distance of $3\sqrt{2}R$ from its centre is:

(1) $\frac{3kq}{\sqrt{6}R}$ (2) $\frac{4kq}{\sqrt{19}R}$
(3) $\frac{4kq}{\sqrt{6}R}$ (4) $\frac{3kq}{\sqrt{19}R}$

35. For a photosensitive surface with threshold frequency ν_0 (in Hz), incident light of frequency $10\nu_0$ ejects electrons with a maximum speed of 6×10^6 m/s. If the frequency is changed to $5\nu_0$, the new maximum speed of the photoelectrons will be:

(1) 2×10^6 m/s
(2) 4×10^6 m/s
(3) 3×10^6 m/s
(4) 9×10^6 m/s

36. A particle of mass 1 mg and charge + 1 μC is projected with speed 50 m/s towards a region of magnetic field 10 Tesla . What is the minimum thickness (*d*) of this region (in m) so that particle returns back to left side.



(1) 1 m
(2) 1.2 m
(3) 0.7 m
(4) 1.50 m

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37. The effective mass of a photon in motion having frequency ν is: (symbols have usual meaning)

(1) $\frac{c}{h\nu}$ (2) $\frac{h}{\nu}$

(3) $h\nu$ (4) $\frac{h\nu}{c^2}$

38. A radioactive nucleus ${}^{A}_{Z}X$ undergoes a series of decays, emitting four α particles and two β^- particles. The final stable nucleus formed is ${}^{206}_{82}Pb$. What are the mass number (A) and atomic number (Z) of the original nucleus ${}^{A}_{Z}X$.

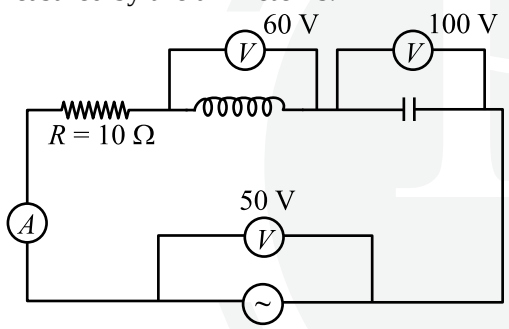
(1) $A = 222, Z = 90$

(2) $A = 238, Z = 92$

(3) $A = 232, Z = 90$

(4) $A = 222, Z = 88$

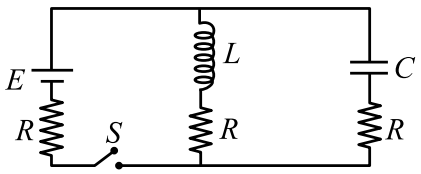
39. In the circuit shown in the figure, the current measured by the ammeter is:



(1) 1 A (2) 2 A

(3) 3 A (4) Zero

40. The switch shown in the circuit is closed at $t = 0$. The current drawn from the battery by the circuit at $t = 0$ and $t = \infty$ are in the ratio:



(1) 2:1 (2) 1:2

(3) 1:1 (4) 1:4

41. If the potential energy of an electron in a hydrogen atom in the first excited state is taken to be zero, the kinetic energy (in eV) of an electron in the ground state will be:

(1) 13.6 eV

(2) 10.2 eV

(3) 3.4 eV

(4) 5.1 eV

42. Two thin plano-convex lenses of equal focal lengths and are in contact are arranged as shown. The ratio of the combined focal lengths (from left to right) is:



(1) 1:2:1 (2) 1:2:3

(3) 1:1:1 (4) 2:1:2

43. A piece of conducting wire of resistance R is cut into $2n$ equal parts. Half of the parts are connected in series to form a bundle and remaining half in parallel to form another bundle. These bundles are then connected such that they give the maximum resistance. The maximum resistance of the combination is:

(1) $\frac{R}{2}\left(1+\frac{1}{n^2}\right)$

(2) $\frac{R}{2}\left(1+n^2\right)$

(3) $\frac{R}{2(1+n^2)}$

(4) $R\left(n+\frac{1}{n}\right)$

44. When ${}_{90}Th^{228}$ gets converted into ${}_{83}Bi^{212}$, then the number of α and β^- particles emitted will respectively be:

(1) $4\alpha, 7\beta^-$ (2) $4\alpha, 1\beta^-$

(3) $8\alpha, 7\beta^-$ (4) $4\beta^-, 4\alpha$

45. Two identical thin rings, each of radius a are placed coaxially at a distance "a" apart. Let charges Q_1 and Q_2 be placed uniformly on the two rings. The work done by external agent in moving a charge Q from the centre of ring of charge Q_2 to the centre of the ring with charge Q_1 is:

(1) zero

(2) $\frac{Q\sqrt{2}}{4\pi\epsilon_0 a}(Q_1+Q_2)$

(3) $\frac{Q(\sqrt{2}-1)}{4\pi\epsilon_0 a\sqrt{2}}(Q_1-Q_2)$

(4) $\frac{Q(\sqrt{2}+1)}{4\pi\epsilon_0 a}(Q_1-Q_2)$

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46. How many grams of sucrose (molar mass = 342g mol^{-1}) should be dissolved in 100 grams H_2O in order to produce a solution with 105°C difference between the freezing point & boiling point temperature?

(K_f = 1.86° C kg/mole) & (K_b = 0.51° C kg/mole)

(1) 34.2 gm (2) 72 gm

(3) 342 gm (4) 460 gm

47. The E_{cell} for the following cell

$Cu(s)|Cu^{2+}(0.01M)||Cu^{2+}(0.04M)|Cu(s)$ is:

(1) $+\frac{RT}{F}\ln(0.5)$

(2) $+\frac{RT}{F}\ln(0.25)$

(3) $-\frac{RT}{F}\ln(0.5)$

(4) $-\frac{RT}{2F}\ln(0.5)$

48. Two reactions R_1 & R_2 have identical pre-exponential factors. Activation energy of R_1 exceeds that of R_2 by 10 kJ mol^{-1}. If k_1 & k_2 are the rate constants for reaction R_1 & R_2 respectively at 300 K, then ln (k_2/k_1) is equal to:

(1) 12 (2) 6

(3) 4 (4) 8

49. Which of the following is an anhydride of nitrous acid?

(1) N_2O_3 (2) NO

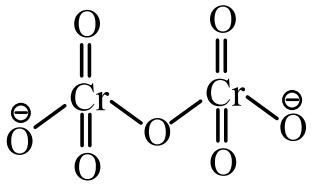
(3) NO_2 (4) N_2O_5

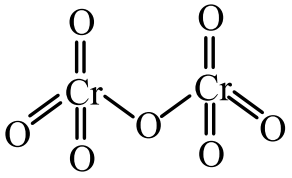
50. The halogen which is solid at room temperature is:

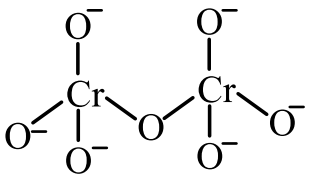
(1) I_2 (2) Br_2

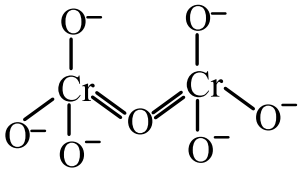
(3) Cl_2 (4) F_2

51. Identify the correct structure of dichromate ion:

(1) 

(2) 

(3) 

(4) 

52. IUPAC name of $Na_3[CoCl(NO_2)_5]$ is:

(1) Sodium chloridopentanitrocobaltate(III)

(2) Sodium cobaltnitrate

(3) Trisodium chloropentanitro cobalt

(4) Pcntanitrocobalt (III) trisodium complex

53. XeF_6 on complete hydrolysis gives;

(1) Xe (2) XeO_2

(3) XeO_3 (4) XeO_4

54. Zr and Hf have almost equal atomic and ionic radii because of:

(1) Diagonal relationship

(2) Lanthanoid contraction

(3) Actinoid contraction

(4) Belong to the same group

55. Match the formulas of oxides given in Column-I with the type of oxides given in Column-II, and mark the correct option.

Column-I		Column-II	
(a)	Pb_3O_4	(i)	Neutral oxide
(b)	N_2O	(ii)	Acidic oxide
(c)	Mn_2O_7	(iii)	Basic oxide
(d)	Bi_2O_3	(iv)	Mixed oxide

Choose the **correct** answer from the options given below:

(1) (a)-(i); (b)-(ii); (c)-(iii); (d)-(iv)

(2) (a)-(iv); (b)-(i); (c)-(ii); (d)-(iii)

(3) (a)-(iii); (b)-(ii); (c)-(iv); (d)-(i)

(4) (a)-(iv); (b)-(iii); (c)-(i); (d)-(ii)

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56. Compounds $[PtCl_2(NH_3)_4]Br_2$ and $[PtBr_2(NH_3)_4]Cl_2$ shows the following type of isomerism:

(1) Hydrate isomerism
(2) Ionisation isomerism
(3) Linkage isomerism
(4) Coordination position isomerism

57. The electronic configuration of Thorium is

(1) $[Rn]6d^5\ 7s^2$
(2) $[Rn]6d^2\ 7s^2$
(3) $[Rn]5f^{10}\ 7s^2$
(4) $[Rn]5f^6\ 7s^2$

58. Which of the following statement (s) are **correct**?

A. In group 15 elements covalent and ionic radii increases down the group but there is small increase from As to Bi.

B. Ionizations enthalpy of group 15 element is less than that of 14 elements, in corresponding period.

C. In group 15 elements, electronegativity decreases down the group with increasing atomic size.

D. In group 15 elements, metallic character increases down the group.

(1) A, B and C (2) Only A and D
(3) A, C and D (4) B, C and D

59. The number of Faradays (F) required to produce 20g of calcium from molten $CaCl_2$ (Atomic mass of Ca = 40 g mol^{-1}) is:

(1) 2 (2) 3
(3) 4 (4) 1

60. Match the items of Column-I and Column-II.

Column-I		Column-II	
(a)	Mathematical expression for rate of reaction	(i)	Rate constant
(b)	Rate of reaction for zero order reaction is equal to	(ii)	Rate law
(c)	Units of rate constant for zero order reaction is same as that of	(iii)	Second^{-1}
(d)	Unit of rate constant for First order reaction	(iv)	Rate of reaction

Choose the **correct** answer from the options given below:

(1) (a)-(ii); (b)-(i); (c)-(iii); (d)-(iv)
(2) (a)-(iii); (b)-(iv); (c)-(i); (d)-(ii)
(3) (a)-(ii); (b)-(i); (c)-(iv); (d)-(iii)
(4) (a)-(i); (b)-(ii); (c)-(iv); (d)-(iii)

61. Fluorine forms only one oxoacid whereas other halogens form several oxoacids. This is mainly because fluorine:

(1) Has the highest electron gain enthalpy
(2) Lacks d-orbitals in the valence shell
(3) Has very high density
(4) Has low electronegativity

62. The yellow colored solution of Na_2CrO_4 changes to orange red on passing CO_2 gas due to the formation of:

(1) CrO_5 (2) CrO_3
(3) $Na_2Cr_2O_7$ (4) Na_3CrO_8

63. Given below are two statements:

Statement-I: The bond dissociation enthalpy of F-F is lower than that of Cl-Cl

Statement-II: Strong lone pair-lone pair repulsions between closely placed fluorine atoms weaken the F-F bond

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct
(2) Both Statement-I and Statement- II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-II is correct but Statement-I is incorrect.

64. The effective atomic number for Rh in $[Rh(H_2O)_6]^{3+}$ (atomic number of Rh is 45) is

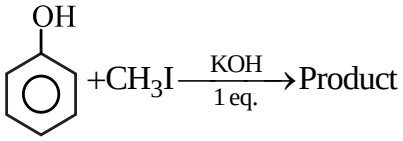
(1) 42 (2) 45
(3) 48 (4) 54

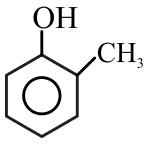
65. The **correct** order of decreasing second ionization enthalpy of Ti (22), V(23), Cr(24), and Mn (25) is:

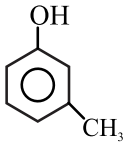
(1) Ti > V > Cr > Mn
(2) Cr > Mn > V > Ti
(3) V > Mn > Cr > Ti
(4) Mn > Cr > Ti > V

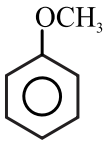
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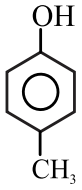
66. What is the final product of the given reaction?



(1) 

(2) 

(3) 

(4) 

67. Given below are two statements:

Statement-I: Fe^{2+} is paramagnetic in nature.

Statement-II: Fe^{2+} has 4 unpaired electron.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

68. Iso propyl alcohol $\xrightarrow[\Delta]{Al_2O_3} B \xrightarrow[H_2SO_4]{Conc.} C \xrightarrow[boil]{water} D$

In this reaction D is:

(1) Iso-propyl alcohol

(2) Propene

(3) n-butyl alcohol

(4) n-propyl alcohol

69. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: $CrCl_3.4H_2O$ gives white ppt with $AgNO_3$.

Reason R: It contains free chloride ion.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

70. Which of the following statements are **correct**?

i. Benzyl halides are more reactive than vinyl and aryl halides.

ii. Vinyl halides are more reactive than alkyl halides.

iii. Aryl halides are less reactive than alkyl halides

iv. Aryl halides are more reactive than benzyl halides.

(1) i & ii (2) i & iii

(3) i, ii & iii (4) iii & iv

71. Chlorination of toluene in the presence of light and heat followed by treatment with aqueous KOH gives

(1) o–cresol (2) m–cresol

(3) p-cresol (4) benzyl alcohol

72. Which of the following are amphoteric oxide?

Mn_2O_7, CrO_3, Cr_2O_3, CrO, V_2O_5, V_2O_4

(1) V_2O_5, Cr_2O_3 (2) Mn_2O_7, CrO_3

(3) CrO, V_2O_5 (4) V_2O_5, V_2O_4

73. The end product (Z) in the given sequence of reaction is:

$$CH_3CH=CHCHO \xrightarrow{NaBH_4} X \xrightarrow[ZnCl_2]{HCl} Y \xrightarrow[(ii)\ H_3O^+]{(i)\ KCN} Z$$

(1) $CH_3CH=CHCH_2COOH$

(2) $CH_3CH_2CH_2COOH$

(3) $CH_3CH=CHCOOH$

(4) $CH_3CH(Cl)CH_2COOH$

74. The order of energy absorbed which is responsible for the color of given complexes is;

A. $[Ni(H_2O)_2(en)_2]^{2+}$

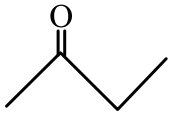
B. $[Ni(H_2O)_4(en)]^{2+}$

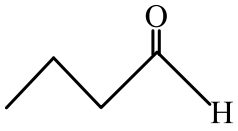
C. $[Ni(en)_3]^{2+}$

(1) B > A > C (2) A > B > C

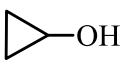
(3) C > B > A (4) C > A > B

75. Compound 'X' C_4H_8O which gives 2, 4-DNP derivative and positive iodoform test is:

(1) 

(2) 

(3) $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-H$

(4) 

[Contd...

76. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Aromatic aldehydes and formaldehyde undergo Cannizaro reaction.

Reason R: Aromatic aldehydes are almost as reactive as formaldehyde.

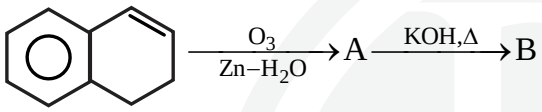
In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

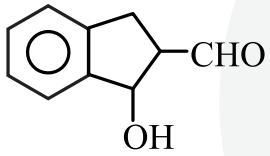
(2) A is false but R is true.

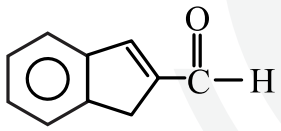
(3) Both A and R are true and R is the correct explanation of A.

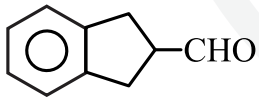
(4) Both A and R are true but R is NOT the correct explanation of A.

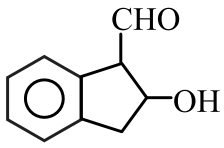
77. 

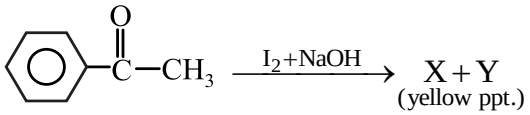
Compound B is:

(1) 

(2) 

(3) 

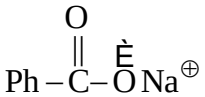
(4) 

78. 

. Identify the final product 'Z'.

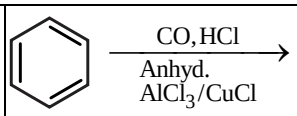
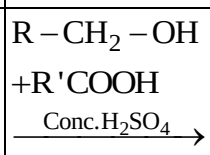
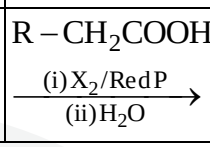
(1) CHI_3

(2) $HC \equiv CH$

(3) 

(4) $H_2C = CH_2$

79. Match Column I with Column II.

Column-I		Column-II	
(A)		(I)	Hell Volhard Zelinsky reaction
(B)		(II)	Gattermann-Koch reaction
(C)		(III)	Haloform reaction
(D)		(IV)	Esterification

Choose the **correct** answer from the options given below.

(1) (A) → (III); (B) → (II); (C) → (I); (D) → (IV)

(2) (A) → (I); (B) → (IV); (C) → (III); (D) → (II)

(3) (A) → (II); (B) → (III); (C) → (IV); (D) → (I)

(4) (A) → (IV); (B) → (I); (C) → (II); (D) → (III)

80. Which of the following is the **correct** order of increasing field strength of ligands to form coordination compounds?

(1) $SCN^- < F^- < CN^- < C_2O_4^{2-}$

(2) $F^- < SCN^- < C_2O_4^{2-} < CN^-$

(3) $CN^- < C_2O_4^{2-} < SCN^- < F^-$

(4) $SCN^- < F^- < C_2O_4^{2-} < CN^-$

81. Niobium (Nb) and ruthenium (Ru) have "x" and "y" number of electrons in their respective 4d orbitals. The value of $x + y$ is

(1) 11 (2) 12

(3) 10 (4) 8

82. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Acetaldehyde and Acetone both react with Tollen's reagent to form silver mirror.

Reason R: Both, aldehydes and ketones contain a carbonyl group.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

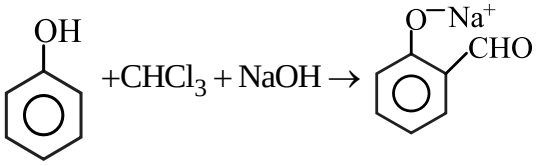
(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

[Contd...

83. In the reaction, the electrophile involved is:



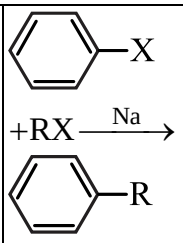
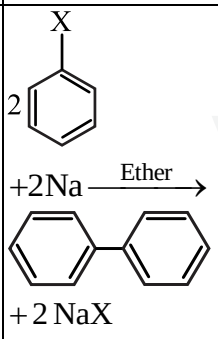
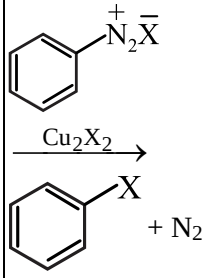
(1) Dichloromethyl cation

(2) Formyl cation $\left(\overset{\oplus}{C}HO\right)$

(3) Dichlorocarbene $\left(:CCl_2\right)$

(4) Dichloromethyl anion $\left(\overset{\ominus}{C}HCl_2\right)$

84. Match the reactions given in Column I with the names given in Column II.

Column-I		Column-II	
(a)		(i)	Fitting reaction
(b)		(ii)	Wurtz Fitting reaction
(c)		(iii)	Finkelstein reaction
(d)	$C_2H_5Cl + NaI \xrightarrow{\text{dry acetone}} C_2H_5I + NaCl$	(iv)	Sandmeyer reaction

Choose the **correct** answer from the options given below.

(1) (a) → (i); (b) → (iii); (c) → (ii); (d) → (iv)

(2) (a) → (iv); (b) → (ii); (c) → (i); (d) → (iii)

(3) (a) → (ii); (b) → (i); (c) → (iv); (d) → (iii)

(4) (a) → (iv); (b) → (i); (c) → (iii); (d) → (ii)

85. $C_6H_6 \xrightarrow[H_2SO_4]{HNO_3} P \xrightarrow{Sn/HCl} Q \xrightarrow[HCl]{NaNO_2} R \xrightarrow{H_3PO_2} S$

The end product S in the given sequence of reaction is:

(1) Benzoic acid

(2) Benzene

(3) Phenol

(4) Chlorobenzene

86. Λ_m^o for NaCl, HCl and NaAc are 126.4, 425.9 and 91.0 S cm^2 mol^{-1} respectively. Calculate Λ^o for HAc.

(1) 135.8 S cm^2 mol^{-1}

(2) 148.6 S cm^2 mol^{-1}

(3) 224.8 S cm^2 mol^{-1}

(4) 390.5 S cm^2 mol^{-1}

87. Which statement is not **correct**?

(1) 1° – Amines form hydrogen bond.

(2) Ethyl amine has higher boiling point than propane.

(3) Methyl amine is more basic than ammonia.

(4) Dimethyl amine is less basic than methyl amine.

88. The mixtures which shows positive deviation from Raoult's law is:

(1) Benzene + Toulene

(2) Acetone + Chloroform

(3) Chloroethane + Bromoethane

(4) Ethanol + Acetone

89. Acidic amino acid is:

(1) Glycine

(2) Valine

(3) Leucine

(4) Glutamic acid

90. For a first order reaction, the rate constant is 6.909 min^{-1}. The time taken for 75% conversion in minutes is:

(1) $\frac{3}{2}\log 2$ (2) $\frac{2}{3}\log 3$

(3) $\frac{2}{3}\log 2$ (4) $\frac{3}{2}\log\frac{3}{4}$

[Contd...

91. Pollen grains of rice and wheat lose their viability:
(1) after 30 hours of their release.
(2) within 30 minutes of their release.
(3) after 3 months of their release.
(4) after 3 days of their release.

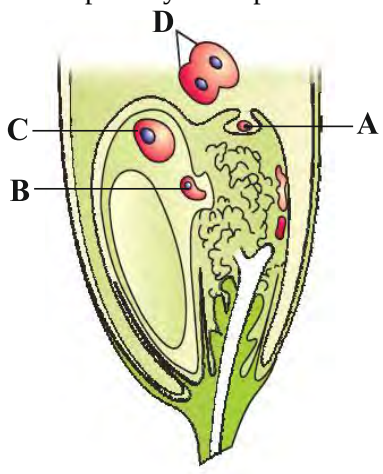
92. In a pond, there were 50 lotus plants last year. Through reproduction, 10 new plants were added. What is the birth rate per lotus per year?
(1) 0.5 offspring per lotus per year
(2) 0.2 offspring per lotus per year
(3) 0.02 offspring per lotus per year
(4) 0.8 offspring per lotus per year

93. Which of the following **correctly** matches the organism with its DNA content?
(1) Bacteriophage $\phi \times 174$ – 48,502 bp
(2) Bacteriophage Lambda – 5,386 nucleotides
(3) *E. coli* – 4.6×10^6 bp
(4) Human DNA (haploid) – 6.6×10^9 bp

94. Which of the following alcoholic beverages is produced **without** distillation of the fermented broth?
(1) Whisky
(2) Brandy
(3) Wine
(4) Rum

95. The vertical distribution of different species occupying different levels in a biotic community is best described as:
(1) trophic level.
(2) stratification.
(3) ecological pyramids.
(4) standing crop.

96. Fusion of which labelled structures results in the formation of primary endosperm nucleus?



(1) B and C (2) A, B and C
(3) A and C (4) A and D

97. Match **List-I** with **List-II**.

List-I (Scientists)		List-II (Contributions)	
(A)	Har Gobind Khorana	(I)	Cell-free system for protein synthesis
(B)	Marshall Nirenberg	(II)	Used polynucleotide phosphorylase enzyme
(C)	Severo Ochoa	(III)	Identified DNA as nuclein
(D)	Friedrich Miescher	(IV)	Synthesised RNA molecules with defined combinations of bases

Choose the *most appropriate* answer from the options given below:
(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-II, C-I, D-IV
(4) A-I, B-III, C-II, D-IV

98. Which of the following is **not** a reason to explain the greater biological diversity in the tropics than temperate regions?
(1) Tropics had more evolutionary time.
(2) Tropical environments are less seasonal and more constant.
(3) There is less solar energy available in the tropics.
(4) Tropical latitudes have remained relatively undisturbed for millions of years.

99. Match **List-I** with **List-II.**

List-I (Cell)		List-II (Feature)	
(A)	Vegetative cell of pollen grain	(I)	Spindle shaped
(B)	Generative cell of pollen grain	(II)	Irregular nucleus, food reserve
(C)	Microspore mother cell	(III)	Undergoes meiosis
(D)	Central cell of embryo sac	(IV)	Contains polar nuclei

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-III, D-IV
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-II, C-IV, D-I

100. In the Meselson-Stahl experiment, after 40 minutes, the DNA was found to be:
(1) 100% Hybrid density.
(2) 50% Hybrid and 50% Light density.
(3) 50% Hybrid and 50% Heavy density.
(4) 100% Light density.

101. Select the organisms that function as biofertilisers.
A. *Rhizobium*
B. *Trichoderma*
C. *Azospirillum*
D. *Anabaena*
E. Baculoviruses
Choose the **correct** answer:
(1) A, B and C only
(2) A, C and D only
(3) B, D and E only
(4) A, C and E only

102. Given below are two statements:
Statement I: Tightly linked genes show very high recombination frequency.
Statement II: Loosely linked genes show low recombination frequency.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

103. Despite occupying about 70% of the surface, the productivity of oceans is approximately (dry weight of organic matter):
(1) 115 billion tons.
(2) 55 billion tons.
(3) 170 billion tons.
(4) 10 billion tons.

104. Hybrid seeds have to be produced every year because:
(1) They lose viability quickly.
(2) Progeny shows segregation of characters.
(3) They are expensive.
(4) Farmers cannot store them.

105. In eukaryotes, RNA polymerase III is responsible for transcribing:
(1) 28S, 18S, 5.8S rRNAs.
(2) hnRNA precursor of mRNA.
(3) tRNA, 5srRNA, and snRNAs.
(4) All types of RNAs.

106. For stable population growth, the number of pre-reproductive individuals is:
(1) much greater than the number of reproductive individuals.
(2) much less than the number of reproductive individuals.
(3) almost equal to the number of reproductive individuals.
(4) equal to the number of post-reproductive individuals.

107. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In honey bees, females are diploid and males are haploid.
Reason (R): Males are produced by parthenogenesis from unfertilized eggs, while females are produced by fertilization.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

108. In a sewage treatment plant, the 'flocs' refer to:
(1) masses of bacteria associated with fungal filaments.
(2) sedimented grit and pebbles.
(3) anaerobic bacteria in the sludge digester.
(4) floating debris removed by filtration.

109. Which of the following extinct animals is **correctly** matched with its country of origin?
(1) Dodo - Africa
(2) Quagga - Russia
(3) Thylacine - Australia
(4) Steller's Sea Cow - Mauritius

110. Which of the following statements regarding the Double Helix Model is **incorrect**?
(1) The backbone is constituted by sugar-phosphate.
(2) The two chains have anti-parallel polarity.
(3) The pitch of the helix is 34 nm.
(4) The distance between two consecutive base pairs in a helix is approximately 0.34 nm.

111. Which of the following are considered as outbreeding devices?
A. Producing unisexual flowers
B. Self-incompatibility
C. Producing bisexual cleistogamous flowers
D. Synchronization of pollen release and stigma receptivity
E. Dioecy
Choose the **correct** answer from the options given below
(1) A, B, C and E only
(2) A, C and D only
(3) B, C and E only
(4) A, B and E only

112. All the genes that are expressed as RNA are referred to as:
(1) Sequence Annotation.
(2) Expressed Sequence Tags (ESTs).
(3) DNA Fingerprinting.
(4) Blind Approach.

113. Greater BOD of waste water indicates:
(1) lower polluting potential.
(2) higher amount of dissolved oxygen.
(3) higher polluting potential.
(4) less organic matter.

114. Decomposition rate is slower if detritus is rich in:
(1) nitrogen and sugars.
(2) lignin and chitin.
(3) sugars and water-soluble substances.
(4) cellulose and nitrogen.

115. The scientists who independently rediscovered Mendel's results in 1900 were:
(1) Watson, Crick, Wilkins.
(2) de Vries, Correns, von Tschermak.
(3) Sutton, Boveri, Morgan.
(4) Hershey, Chase, Griffith.

116. A "true-breeding" line is one that:
(1) shows stable trait inheritance and expression for several generations after undergoing self pollination.
(2) is always heterozygous.
(3) is produced by cross-pollination between unrelated plants.
(4) produces offspring that always differ from the parent.

117. Given below are two statements:
Statement I: The *lac* operon has a monocistronic structural gene.
Statement II: The *y* gene in the *lac* operon codes for permease.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Both Statement I and Statement II are incorrect.
(4) Statement I is incorrect but Statement II is correct.

118. In *Verhulst-Pearl Logistic Growth***,** the carrying capacity (K) represents:
(1) the minimum number of individuals a habitat can support.
(2) the maximum number of individuals a habitat can support.
(3) the intrinsic rate of increase.
(4) the reproductive fitness of the organism or population.

119. Which of the following is **not** a characteristic of wind-pollinated flowers?
(1) Well-exposed stamens
(2) Large feathery stigma
(3) Single ovule in each ovary
(4) Large, colorful flowers with nectar

120. Which of the following human chromosomes has the fewest number of genes and the highest number of genes, respectively?
(1) Chromosome 1 and Chromosome 21
(2) Chromosome 21 and Chromosome 1
(3) Chromosome Y and Chromosome 1
(4) Chromosome Y and Chromosome X

121. The gases produced in an anaerobic sludge digester of a sewage treatment plant are:
(1) Methane, Oxygen and H_2S.
(2) Methane, H_2S and CO_2.
(3) CO_2, SO_2 and Nitrogen.
(4) Methane, Nitrogen and Oxygen.

122. Identify the **incorrect** statement about ecosystems.
(1) Crop fields and an aquarium may be considered as man-made ecosystems.
(2) The decomposers in pond ecosystem are the fungi, bacteria and flagellates.
(3) A constant input of solar energy is the basic requirement for any ecosystem.
(4) Deep sea hydro-thermal ecosystem does not follow second laws of thermodynamics.

123. Histones are rich in which basic amino acid residues?
(1) Histidine and Arginine
(2) Lysine and Arginine
(3) Leucine and Arginine
(4) Leucine and Methionine

124. Match **List I** with **List II**

List-I (Term)		**List-II (Description)**	
(A)	Incomplete dominance	(I)	Physical association of genes
(B)	Linkage	(II)	Heterozygous condition shows intermediate expression not resembling either parent
(C)	Pleiotropy	(III)	Single gene affecting multiple phenotypes
(D)	Polygenic inheritance	(IV)	Multiple genes affecting a single phenotype

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-I, C-II, D-IV

125. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Transcription and translation can be coupled in bacteria.
Reason (R): In bacteria, the mRNA does not require processing and there is no separation of cytosol and nucleus.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

126. Identify the **incorrectly** matched pair.
(1) Cyclosporin A - *Trichoderma polysporum*
(2) Statins - *Monascus purpureus*
(3) Streptokinase - *Streptococcus* sp.
(4) Roquefort cheese - *Propionibacterium sharmanii*

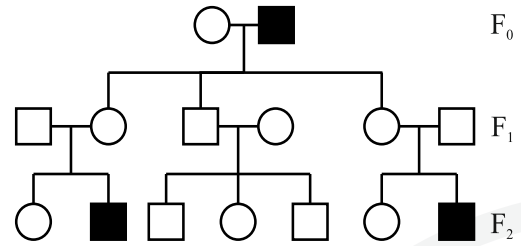
127. Given below are two statements:
Statement I: The number of fungi species in the world is more than the combined total of the species of fishes, amphibians, reptiles, and mammals.
Statement II: Conventional taxonomic methods are highly suitable for identifying microbial species.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Both Statement I and Statement II are incorrect.
(4) Statement I is incorrect but Statement II is correct.

128. Select the **correct** statements regarding Klinefelter's syndrome.
A. The karyotype is 47, XXY.
B. The individual shows overall feminine development.
C. Gynaecomastia is commonly observed.
D. Affected individuals are sterile.
Choose the **correct** answer from the options given below:
(1) A, B and C only
(2) A, C and D only
(3) B, C and D only
(4) A, B and D only

[Contd...

129. If the coding strand sequence is 5'-TACGTAC-3', what is the transcribed mRNA sequence?
(1) 3'-ATGCATG-5'
(2) 5'-UACGUAC-3'
(3) 5'-AUGCAUG-3'
(4) 3'-UACGUAC-5'

130. Study the pedigree chart below. What is the most likely mode of inheritance of trait?



■ : Affected male
□ : Unaffected male
◯ : Unaffected female
(1) X-Linked Dominant
(2) Autosomal Dominant
(3) Y-Linked
(4) X-Linked Recessive

131. Given below are two statements:
Statement I: The female mosquito is considered a true parasite because it needs human blood for reproduction.
Statement II: Human liver fluke is a trematode parasite that depends on two intermediate hosts to complete its life cycle.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Both Statement I and Statement II are incorrect.
(4) Statement I is incorrect but Statement II is correct.

132. Given below are two statements:
Statement I: The exine of pollen grains is made up of pectin and cellulose, which is resistant to high temperatures and strong acids.
Statement II: Sporopollenin is absent at the germ pores in a pollen grain.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

133. Identify the **incorrect** statement.
(1) In bacteria, 23S rRNA acts as a ribozyme.
(2) A ribosome consists of structural RNAs and about 80 different proteins.
(3) Translation of mRNA into protein begins when the large ribosomal subunit encounters the mRNA.
(4) Untranslated regions (UTRs) are required for an efficient translation process.

134. Which of the following is **not** a criteria for a molecule to act as genetic material?
(1) It should be able to generate its replica.
(2) It should be chemically and structurally unstable.
(3) It should provide scope for slow changes (mutation).
(4) It should be able to express itself in the form of 'Mendelian Characters'.

135. Which of the following are features of the Genetic Code?
A. UAA codes for methionine and act as initiator codon
B. The codon is read in mRNA in a contiguous fashion.
C. The code is degenerate.
D. The code is nearly universal.
E. AUG, UAG, UGA are stop terminator codons.
Choose the **correct** answer from the options given below:
(1) A, C, D only
(2) A, B, C only
(3) B, C, D only
(4) A, D, E only

136. Identify the **correct** statement regarding the separation of DNA fragments by Agarose Gel Electrophoresis:
(1) DNA fragments move towards the cathode because they are negatively charged.
(2) The matrix used is agarose, which is a synthetic polymer extracted from sea weeds.
(3) The separated DNA fragments can be visualized as bright orange colored bands after staining with Ethidium bromide and exposure to UV light.
(4) Larger DNA fragments move farther away from the loading wells due to the sieving effect of the gel.

137. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Wombat	(I)	Fungal pathogen
(B)	Lemur	(II)	Placental mammal
(C)	*Microsporum*	(III)	Australian marsupial
(D)	*Plasmodium*	(IV)	Protozoan pathogen

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-II, C-IV, D-I

138. India was among the first countries to initiate 'family planning' programmes in:
(1) 1951 (2) 1971
(3) 2000 (4) 1981

139. Arrange the following events in the **correct** chronological order regarding the origin of life on Earth:
A. Formation of the first cellular forms of life.
B. Synthesis of simple organic compounds (Monomers) like amino acids.
C. Appearance of the first non-cellular forms of life (Giant molecules like RNA, Proteins).
D. Formation of complex organic molecules (Polymers).
Choose the **correct** answer from the options given below:
(1) (B)→ (D) → (C) → (A)
(2) (D) → (B) → (C) → (A)
(3) (B) →(C) → (D) → (A)
(4) (C) → (A) → (B) → (D)

140. Which of the following was the first human hormone produced by recombinant DNA technology?
(1) Estrogen
(2) Insulin
(3) Thyroxine
(4) Progesterone

141. Consider the following statements (A-E) regarding oogenesis and choose the **incorrect** ones.
A. It is initiated during the embryonic development stage.
B. No more oogonia are formed or added after birth.
C. The primary oocyte completes its first meiotic division within the secondary follicle.
D. The first meiotic division is an equal division.
E. At puberty, only 60,000-80,000 primary follicles are left in each ovary.
Choose the **correct** ones:
(1) A, B, and C
(2) A, B, and E
(3) B, C, and D
(4) C and D only

142. Bt toxin is produced by a bacterium called:
(1) *Bacillus anthracis*
(2) *Bacillus thuringiensis*
(3) *Bacillus subtilis*
(4) *Agrobacterium tumefaciens*

143. Which of the following drugs is obtained from the plant whose one flowering branch is shown below?



(1) Morphine (2) Cocaine
(3) Hashish (4) Barbiturates

144. The first transgenic cow, Rosie, produced human protein-enriched milk containing:
(1) Human alpha-lactalbumin
(2) Human beta-globin
(3) Human insulin
(4) Human alpha-1-antitrypsin

[Contd...

145. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The principle of vaccination is based on the property of 'memory' of the immune system.
Reason R: Vaccines generate memory B and T-cells that recognize the pathogen quickly on subsequent exposure.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

146. What is the function of the "foetal ejection reflex" in human parturition?
(1) To trigger the release of oxytocin from the maternal anterior pituitary.
(2) To provide nutrition to the developing embryo.
(3) To stop the menstrual cycle during pregnancy.
(4) To trigger the release of oxytocin from the maternal posterior pituitary.

147. Which of the following is a severe complication often resulting from untreated Pelvic Inflammatory Diseases (PIDs) caused by Sexually Transmitted Infections?
(1) Ectopic pregnancies
(2) *Erythroblastosis foetalis*
(3) Swelling in genital region
(4) Polyembryony

148. Match **List-I** with **List-II.**

List-I (PCR Step/Component)		**List-II (Feature/Function)**	
(A)	Denaturation	(I)	Synthesis of DNA using nucleotides at 72°C
(B)	Annealing	(II)	Separation of the two strands of DNA by high heat
(C)	Extension	(III)	Isolation source of thermostable DNA polymerase
(D)	*Thermus aquaticus*	(IV)	Hybridisation of two sets of primers

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I D-II
(2) A-II, B-IV, C-III, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-IV, C-I, D-III

149. Which of the following is the **correct** chronological sequence of duct systems starting from the seminiferous tubules?
(1) Rete testis → Vasa efferentia →Epididymis →Vas deferens
(2) Vasa efferentia →Rete testis →Epididymis →Vas deferens
(3) Rete testis →Epididymis →Vasa efferentia →Vas deferens
(4) Seminiferous tubules → Rete testis →Vas deferens →Epididymis

150. The "good humor" hypothesis of health was disproved by the discovery of blood circulation by:
(1) Hippocrates
(2) Indian Ayurveda system
(3) William Harvey
(4) M.S. Swaminathan

151. Choose **correctly** matched pair.
(1) Elution → Cutting DNA into fragments
(2) Downstream processing → Initial biosynthetic stage
(3) Bioreactor → Small volume culture vessel
(4) Transformation → Introduction of DNA into a host cell

152. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Non-medicated IUD	(I)	Multiload 375
(B)	Hormone releasing IUD	(II)	Lippes loop
(C)	Copper releasing IUD	(III)	Saheli
(D)	Oral Pill	(IV)	LNG-20

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-IV, C-III, D-I

153. In the geological time scale representing the history of vertebrates, which period lies immediately between the Devonian and the Permian periods?
(1) Triassic (2) Jurassic
(3) Silurian (4) Carboniferous

[Contd...

154. A Hominid fossil discovered in Java in 1891, now extinct, having cranial capacity of about 900 cc was:
(1) *Neanderthal man* (2) *Homo sapiens*
(3) *Australopithecus* (4) *Homo erectus*

155. Choose **correctly** matched pair.
(1) Trophoblast → Inner group of cells attached to blastocyst
(2) Morula → 8 to 16 blastomeres
(3) Placenta → Secretes only relaxin
(4) Endometrium → Middle thick layer of smooth muscle

156. Grafted kidney may be rejected in a patient is primarily due to:
(1) Cell-mediated immune response
(2) Passive immune response
(3) Innate immune response
(4) Humoral immune response

157. Consider the following statements:
Statement I: Amniocentesis is a legally banned procedure for sex determination.
Statement II: In India laws prohibit legal adoption and it is as yet, one of the best methods for couples looking for parenthood.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

158. Which of the following statement is **incorrect** about *Agrobacterium tumifaciens*?
(1) Modified Ti plasmid from *Agrobacterium tumifaciens* used for gene transfer is not pathogenic to plant cell.
(2) It transforms normal plant cells into tumor cells.
(3) It is used to deliver gene of interest in both prokaryotic as well as eukaryotic host cells.
(4) It delivers 'T-DNA' into plant cell.

159. Identify the factors that affect Hardy-Weinberg equilibrium:
A. Gene migration
B. Genetic drift
C. Mutation
D. Genetic recombination
E. Random mating
(1) A, B, and C only
(2) A, B, C, D, and E
(3) B, C, and E only
(4) A, B, C and D only

160. Match **List-I** with **List-II** and select the **correct** combination.

List-I		List-II	
(A)	Gene therapy	(I)	Antigen-antibody interaction
(B)	Gene cloning	(II)	To obtain identical copies of a particular DNA molecule
(C)	ELISA	(III)	Replacement of a faulty gene by a normal healthy gene
(D)	Genetically engineered Human Insulin	(IV)	*E. coli*

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

161. A 25-year-old woman has a regular 28-day menstrual cycle. On which day is the LH surge most likely to induce the rupture of the Graafian follicle?
(1) Day 1
(2) Day 5
(3) Day 14
(4) Day 28

162. Which of the following is a **correct** statement?
(1) All STIs are completely curable.
(2) AIDS are completely curable.
(3) Sterilization is a highly reversible method.
(4) MTP is safe in the first trimester.

163. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Elephantiasis	(I)	*Haemophilus influenzae*
(B)	Pneumonia	(II)	*Wuchereria bancrofti*
(C)	Cancer	(III)	Viral disease
(D)	AIDS	(IV)	Radiotherapy

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-I, C-IV, D-III

[Contd...

164. Which of the following statements is **incorrect**?

(1) A stirred-tank reactor is usually cylindrical or with a curved base to facilitate the mixing of the reactor contents.

(2) Bio-reactors have an agitator system, an oxygen delivery system and no foam control system.

(3) A bio-reactor provides optimal growth conditions for achieving the desired product.

(4) Most commonly used bio-reactors are of stirring type.

165. Identify the **correct** statements (A-E) regarding Malaria:

A. Malignant malaria is caused by *Plasmodium vivax*.

B. The infectious form of *Plasmodium* for humans is the sporozoite.

C. The P*lasmodium* reproduces sexually in the liver cells.

D. Rupture of RBCs releases a toxic substance called haemozoin.

E. The female *Anopheles* mosquito is the vector and secondary host for malaria.

(1) A, B, and C

(2) A, B, D, and E

(3) B, D, and E only

(4) B and D only

166. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Thorns of *Bougainvillea* and tendrils of *Cucurbita*	(I)	Transfer of genes into or out of population
(B)	Eyes of octopus and mammals	(II)	Change in gene frequency by chance.
(C)	Genetic drift	(III)	Homologous organs
(D)	Gene flow	(IV)	Analogous organs

Choose the **correct** answer from the options given below:

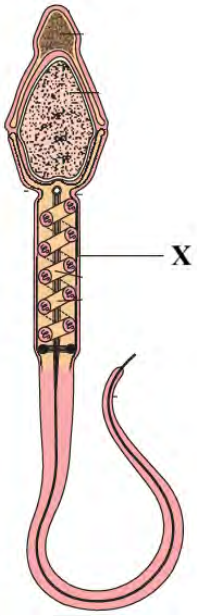
(1) A-I, B-II, C-III, D-IV

(2) A-III, B-IV, C-II, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-IV, B-III, C-I, D-II

167. Which of the following **correctly** represent the function of labelled **X** in the given diagram of human sperm?



(1) Contains enzymes for fertilization

(2) Contains haploid nucleus

(3) Possesses numerous mitochondria for energy to facilitate motility

(4) Acts as a sensory organ for finding the ovum

168. Choose **correct** statement from the options given below.

(1) Malignant tumor cells exhibit contact inhibition.

(2) Patients who have undergone surgery are given cannabinoids to relieve pain.

(3) Benign tumors show metastasis.

(4) Heroin slows down body functions.

169. Which of the following is a **correct** statement?

(1) Traditional hybridization leads to multiplication of only desirable genes.

(2) At present, about 30 recombinant therapeutics have been approved for human-use the world over.

(3) Recombinant DNA technology allows only the introduction of undesirable genes.

(4) Most restriction enzymes are isolated from viruses.

170. Which of the following is **correct** statement?

(1) The universe is 13.8 billion years old.

(2) Earth was formed 4.5 million years ago.

(3) Life appeared 500 billion years after earth's formation.

(4) Louis Pasteur proved spontaneous generation**.**

[Contd...

171. Given below are two statements:
Statement I: The process of release of sperm from the seminiferous tubules is called spermiogenesis.
Statement II: After spermiogenesis, sperm heads become embedded in the Sertoli cells.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

172. Which of the following statements **correctly** describes the fate of a foreign DNA fragment transferred into a host organism?
A. It typically fails to multiply if it remains in the cytoplasm.
B. It replicates autonomously independent of the host genome.
C. It can multiply and be inherited if it integrates into the host chromosome.
D. It utilizes the host's 'origin of replication' after integration.
Choose the **correct** answer:
(1) A and B only (2) B and C only
(3) A, C, and D only (4) A, B, C, and D

173. The theory stating that "Ontogeny repeats Phylogeny" was disapproved by:
(1) Ernst Haeckel
(2) Charles Darwin
(3) Karl Ernst von Baer
(4) Alfred Wallace

174. Transgenic plants are the ones:
(1) Grown on artificial medium after hybridisation in the field.
(2) Produced after protoplast fusion in artificial medium.
(3) Produced by somatic embryo on artificial medium.
(4) Generated by introducing foreign DNA into a cell and regenerating plant from that cell.

175. DNA fragments generated by the restriction endonuclease in a chemical reaction can be separated by:
(1) Restriction mapping
(2) Centrifugation
(3) Polymerase chain reaction
(4) Electrophoresis

176. A couple is unable to conceive despite 2 years of unprotected sexual cohabitation. Which technique involves the transfer of an embryo with more than 8 blastomeres into the uterus?
(1) ZIFT (2) IUI
(3) GIFT (4) IUT

177. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Labia majora	(I)	A membrane that often partially covers the opening of the vagina
(B)	Labia minora	(II)	Fleshy folds of tissue, which extend down from the mons pubis and surround the vaginal opening
(C)	Mons pubis	(III)	Paired folds of tissue under the labia majora
(D)	Hymen	(IV)	Cushion of fatty tissue covered by skin and pubic hair

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-II, B-III, C-I, D-IV
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-IV, C-III, D-I

178. After entering the host's body, HIV enters into which specific cells, replicates, and destroys them to release progeny viruses?
(1) Erythrocytes (2) Helper T-cells (T_H)
(3) Platelets (4) B-cells

179. Which of the following conditions can **not** be diagnosed using amniocentesis?
(1) Down Syndrome (2) Haemophilia
(3) Jaundice (4) Sickle-cell anaemia

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: When a particular restriction enzyme cuts strand of DNA, overhanging stretches or sticky ends are formed.
Reason R: All restriction enzymes cut the strand of DNA a little away from the centre of the palindromic site.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

■■■

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